

$$1a) \frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z$$

$$\frac{p_1}{\rho g} + \frac{3^2}{2g} + 0 = 0 + \frac{V_2^2}{2g} + 0.75$$

$$V_1 A_1 = V_2 A_2$$

$$3 \left(\frac{0.05}{2} \right)^2 \pi = V_2 \left(\frac{0.025}{2} \right)^2 \pi$$

$$V_2 = 12 \text{ m s}^{-1}$$

$$\frac{p_1}{\rho g} + \frac{3^2}{2g} = \frac{12^2}{2g} + 0.75$$

$$p_1 = 1000 \times 9.81 \left(\frac{12^2 - 3^2}{2 \times 9.81} + 0.75 \right)$$

$$= 74.8575 \text{ kPa}$$

$$b) \sum F_x = \dot{m} (V_{2x} - V_{1x})$$

$$\dot{m} = 3 \left(\frac{0.05}{2} \right)^2 \pi \times 1000 = \frac{15}{8} \pi$$

$$\dot{m} (V_{2x} - V_{1x}) = p_1 A_1 + F_x$$

$$\frac{15}{8} \pi (12 \cos 45^\circ - 3) = 74.8575 \times 10^3 \left(\frac{0.05}{2} \right)^2 \pi + F_x$$

$$F_x = -114.6713832$$

$$\approx -114.67 \text{ N}$$

$$1b) \sum F_y = m(v_{2y} - v_{1y})$$

$$v_{2y} = 12 \sin 45^\circ$$

$$v_{1y} = 0$$

$$F_y = m(v_{2y})$$

$$= \frac{15}{8} \pi (12 \sin 45^\circ)$$

$$= 49.98243305$$

$$\approx 49.98 \text{ N}$$

$$F = \sqrt{F_x^2 + F_y^2}$$

$$= \sqrt{(-114.67)^2 + 49.98^2}$$

$$= 125.0910458 \text{ N}$$

$$\approx 125.09 \text{ N}$$

$$\theta = \arctan\left(\frac{F_y}{F_x}\right)$$

$$= \arctan\left(\frac{49.98}{114.67}\right)$$

$$= 23.55120187^\circ$$

$$\approx 23.6^\circ$$

2) Taking moments about O:

$$F(50) = 6\left(\frac{15}{2}\right)$$

$$F = 0.9 \text{ N}$$

$$F = \dot{m}(V_{1x} - V_{2x})$$

$$0.9 = \dot{m}V_{1x} - 0$$

$$0.9 = \dot{m}V_{1x}$$

$$0.9 = \rho Q V_{1x}$$

$$0.9 = 1000 Q V_{1x}$$

$$0.9 = 1000 V_{1x}^2 A$$

$$0.9 = 1000 \left(\frac{0.01}{2}\right)^2 \rho V_{1x}^2$$

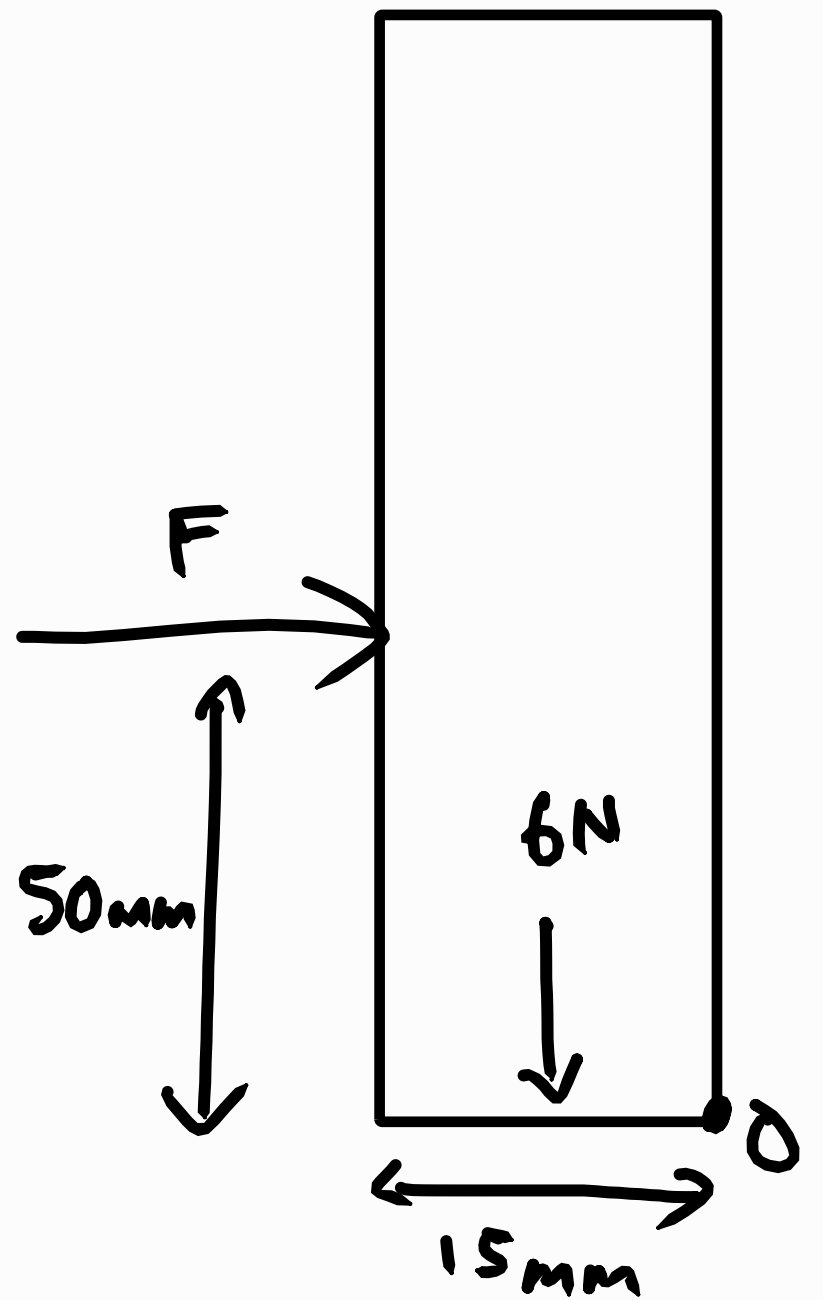
$$V_{1x} = 3.385137501 \text{ m s}^{-1}$$

$$Q = V_{1x} A$$

$$= \left(\frac{0.01}{2}\right)^2 (3.385137501)$$

$$= 2.658680776 \times 10^{-4}$$

$$\approx 2.66 \times 10^{-4} \text{ m}^3 \text{ s}^{-1}$$



$$3) \sum F_y = \dot{m}(v_{1y} - v_{2y})$$

$$\nabla \nearrow 30 \text{ ms}^{-1}$$

$$0.1 + mg = \dot{m}(30 - 30 \cos 30^\circ)$$

$$mg = 1.23 \left(\frac{10 \times 10^{-2}}{2} \right)^2 \pi (30) (30 - 30 \cos 30^\circ) - 0.1$$

$$mg = 1.064823058$$

$$m = \frac{1.065}{9.81}$$

$$= 0.1085446542$$

$$\approx 0.109 \text{ kg}$$

$$4) \sum F_y = \dot{m}(v_{2y} - v_{1y})$$

$$\sum F_y = 1.5 \times 9.81$$

$$= 14.715 \text{ N}$$

$$\dot{m} = \rho V A$$

$$= 1000(10) \left(\frac{20 \times 10^{-3}}{2} \right)^2 \pi$$

$$= \pi \text{ kg s}^{-1}$$

$$\sum F_y = \dot{m}(v_{2y} - v_{1y})$$

$$-14.715 = \pi(-v_{1y})$$

$$v_{1y} = 4.683929975$$

$$\approx 4.68 \text{ ms}^{-1}$$

4) Bernoulli: between 0 and 1,

$$\frac{\cancel{P_0}}{\cancel{\rho g}} + \frac{V_0^2}{2g} + z_0 = \frac{\cancel{P_1}}{\cancel{\rho g}} + \frac{V_1^2}{2g} + z_1$$

$$\frac{10^2}{2g} = \frac{4.68^2}{2g} + z_1$$

$$z_1 = 3.978634046$$

$$\approx 3.98 \text{ m}$$